

Question (01):-

The main problem addressed by paper is that, the modern generative models can create image that look very realistic but they may not correctly understand and produce the projected geometry of real world scenes.

The paper study error in perspective, vanishing points, textures and image signals but this paper detects generated image using geometric feature instead of raw pixels.

Question (2):-

The main idea of paper is that, AI generated image can be detected through their geometric mistakes. The three geometric properties which are studied by authors are,

1) Object - Shadow Relationship:

checking whether shadows are consistent with their objects.

2) Perspective Fields:

checking whether perspective and

vanishing point information is correct.

3) Line Segment:-

checking how lines are arranged and whether they follow paper perspective.

The important contribution is that, these classifiers do not directly use the original image pixels. They use desired geometric information.

Question (03):-

This method collects real image and generates similar image using different image generators. ResNet-50 signal based preconditions then used to separate easy cases from difficult cases. The author's mainly study image, that are difficult for the normal detector. After that three type of geometric information are extracted:-

1) object-shadow:-

An object-shadow deficit method is used to obtain and shadow masks. A ResNet-50 classifies them studies

the relationship between object and their shadow. It can detect mistakes such as ~~mirror~~ incorrect shadow direction or length

2) Perspective Fields:-

A pretrained perspective fields model is used to obtain perspective information such as gravity and latitude related maps. A ResNet-50 classifier uses this information to identify incorrect perspective.

3) Line Segment:-

The DeepSLD method studies the set of line segments and their arrangements to find geometric inconsistencies.

Finally Grid Ground CAM is used to show which pairs of geometric information the classifier are focusing on.

Question (04):-

The paper used both indoor and outdoor real image. Datasets are:-

- 1) LSON $\rightarrow$ indoor
- 2) BDD 100k $\rightarrow$ outdoor driving scenes.

- 3) mapillary videos $\rightarrow$ outdoor Street Scenarios.
- 4) Recent Akinetemp test set $\rightarrow$ recent indoor and outdoor images collected from online resources.

Gen

Generative models used are:-

- 1) Stable Diffusion XL V2.0
- 2) Kandinsky - V3
- 3) DeepFloyd - V1.0
- 4) PixArt $\rightarrow$ V1.0

For composition, ResNet-50 Signal-based prequalification was used and paper also compares the geometry based methods with simple statistics feature using Logistic regression. These statistical features include line counts and lengths object. Shadow counts and changes in gravity information.

Question 05:-

in my opinion the method generates reasonably well, because the geometry based classifiers trained using one generator can also detect image from other generators. The paper reports such as stable Diffusion

XL, PixArt-2, DALL-E 3, and Adobe Firefly. However, it is not perfect. Deployed in outdoor or combined scenes can also be challenging.

This method may fail when a future generative becomes better at producing correct geometry. These too the method is ~~used~~ useful but should not be considered a perfect detector for every future generator.

Question (06) :-

it opens several interesting research questions. First researchers can investigate whether future generative models can learn correct projective geometry instead of producing small geometric mistakes. Another question is whether geometric aware loss function can add to image generation method to improve perspective, lines and shadows. Researchers can also study whether geometry-base detectors can remain

effective against future generations. Instead of only using geometry to detect false images future users could use geometric information directly to improve the image generation process.